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Modern Delay-Tolerant Email

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June 27, 2026

Submitted in partial fulfilment of the requirements for the degree of
M.Sc. Computer Science - Future Networked Systems

Abstract

Abstract content

Acknowledgements

I would like to thank my supervisor, Professor Stephen Farrell. Under his supervision, I was able to successfully complete this dissertation. His inspiration and suggestions regarding the project's direction were essential for the progress of this research.

I am also grateful to my family and friends for their support and encouragement throughout this process.

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1 Introduction

1.1 Introduction

Delay/Disruption-Tolerant Networking (DTN) and Internet email are both store-and-forward communication systems that enable asynchronous message delivery across heterogeneous networks (1, 2). DTN generalizes the store-and-forward concept as a network-layer or overlay architecture for challenged environments, whereas email is a specific application-layer service built on top of relatively well-connected IP networks.(1, 2)

1.2 Delay/Disruption-Tolerant Networking

1.2.1 Concept and Motivation

DTN targets "challenged" networking environments characterized by intermittent connectivity, long or variable delays, high error rates, and constrained resources such as power and storage.(1, 3) In such environments, classical TCP/IP assumptions about continuous end-to-end paths and low round-trip time do not hold, which requires a different architectural approach.(1)

Typical DTN scenarios include deep-space communication, rural and remote connectivity, ad hoc vehicular and mobile networks, and disaster-response deployments where infrastructure is damaged or overloaded.(1, 3) The primary objective is to provide eventual data delivery despite disruptions, using persistent storage and opportunistic forwarding across time-varying connectivity graphs.(1, 4)

1.2.2 DTN Architecture and Bundle Layer

The canonical DTN architecture introduces a new overlay protocol layer, commonly called the bundle layer, which resides above region-specific transports (e.g., TCP, UDP, Licklider Transmission Protocol) and below applications.(1, 3) Applications generate protocol data units called bundles, which encapsulate payload data together with endpoint identifiers, lifetime, priority, and optional security blocks.(1, 5)

Each DTN node maintains persistent storage for bundles and performs a store-carry-forward operation: bundles are stored locally, carried while the node is mobile or idle, and forwarded when a suitable contact becomes available.(1, 6) Convergence-layer adapters map bundle operations onto specific underlying transports (for example, the DTN TCP Convergence-Layer Protocol) so that the bundle layer is insulated from link heterogeneity.(5)

1.2.3 DTN Nodes, Regions, and Security

The architecture partitions the overall system into regions, such as terrestrial IP networks, satellite segments, and deep-space links, which may employ different lower-layer protocols and routing schemes.(3) The bundle layer provides interoperability across these regions using endpoint identifiers and late binding, allowing applications to communicate without being aware of underlying heterogeneity.(3, 6)

Security is addressed using dedicated bundle security mechanisms that provide authentication, integrity protection, and confidentiality over multi-hop, disruption-prone paths.(3) Because DTN nodes may store data for long periods and may replicate messages to improve delivery probability, the security design also considers access control, resistance to resource exhaustion, and key management across regions.(4, 6)

1.2.4 Routing, Scheduling, and Buffer Management

Routing in DTNs differs fundamentally from traditional IP routing because an end-to-end path between source and destination may not exist at any given time.(4, 6) Instead, routing algorithms operate over time-varying or probabilistic contact graphs, taking into account predicted or observed node mobility and contact opportunities.(6)

Existing DTN routing schemes can be broadly classified by the degree of replication and the information they exploit: epidemic and multi-copy protocols maximize delivery probability at the cost of bandwidth and buffer usage, while single-copy or quota-based schemes trade delivery ratio for reduced overhead.(4, 6) Since contacts are intermittent and buffers are limited, joint scheduling and buffer management policies are required to decide which bundles to transmit or drop according to priorities, deadlines, and resource constraints.(6)

1.3 Internet Email Architecture

1.3.1 Service Model and Roles

Internet email is an asynchronous application-layer service that delivers messages between users identified by email addresses of the form `local-part@domain`.(2, 7) The system is decomposed into functional roles including Message User Agents (MUAs), Message Submission Agents (MSAs), Message Transfer Agents (MTAs), Message Delivery Agents (MDAs), and Mail Access Agents.(2)

MUAs (such as desktop clients or webmail front-ends) are responsible for message composition, display, and user interaction, but they offload submission, relay, and storage to server-side agents.(2) MSAs accept messages from MUAs, apply initial checks, and hand them to MTAs, which relay messages between domains until they reach an MDA that deposits them into per-user mailboxes.(2, 7)

1.3.2 Protocol Suite: SMTP, POP, IMAP, MIME

The Simple Mail Transfer Protocol (SMTP) defines the submission and relay of messages between MSAs and MTAs, including envelope commands and response codes for reliable transfer (7). Originally designed for clear text operation and implicit trust between servers, SMTP has been extended with authentication and encryption mechanisms such as SMTP AUTH and STARTTLS.(7)

For mailbox access, the Post Office Protocol version 3 (POP3) supports simple download-and-delete retrieval, while the Internet Message Access Protocol (IMAP) offers folder hierarchies, server-side search, and synchronization across multiple clients and devices.(8) At the message format level, the Internet Message Format standard and Multipurpose Internet Mail Extensions (MIME) specify headers, structure, multiple body parts, and attachments, enabling rich content and internationalization.(9)

1.3.3 Security and Anti-Abuse Mechanisms

Traditional email architecture did not include end-to-end confidentiality, strong sender authentication, or robust abuse controls, making it vulnerable to spoofing, spam, and phishing (2). In practice, deployments mitigate these weaknesses by combining transport-layer security (TLS), server and user authentication, and extensive content-based filtering, often using machine learning to classify spam and malicious messages.(2, 10)

Domain-level authentication and policy mechanisms such as Sender Policy Framework (SPF), DomainKeys Identified Mail (DKIM), and DMARC enable domains to assert which hosts may send mail on their behalf and to specify handling rules for authentication failures.(2) Additional secure application architectures have been proposed that overlay existing email infrastructure with stronger identity, integrity, and legal guarantees, while preserving compatibility with standard protocols.(10)

1.4 Comparison of DTN and Email Architectures

Both DTN and Internet email rely on asynchronous, store-and-forward operation, with intermediate nodes temporarily buffering data before forwarding it to the destination.(1, 2) However, DTN is conceived as a general-purpose network-layer architecture for environments with extreme delay and disruption, while email is a specific application protocol suite designed for relatively stable IP networks (1, 7).

DTN introduces a unified bundle layer to integrate heterogeneous underlying networks and explicitly models time-varying connectivity and contact opportunities (1, 3). On the contrary, the email architecture builds on existing IP transport and focuses on application-level roles, standardized message formats, and security mechanisms for addressing and content-level

1.5 Bundle Protocol in Interplanetary DTN

The Bundle Protocol (BP) is the core network protocol of Delay/Disruption-Tolerant Networking (DTN), designed to support long-delay and intermittently connected paths using store-and-forward transfer of application data units called bundles (11). In interplanetary scenarios, BP is typically deployed as an overlay over one or more convergence layers (such as TCP, LTP, or space link protocols), providing persistent storage and late binding of next hops so that disruptions that would break conventional end-to-end transports do not prevent eventual delivery (11). This overlay role allows BP to act as a unifying backbone across heterogeneous regional networks, particularly where deep-space links have radically different characteristics from local terrestrial segments (11).

1.5.1 Interplanetary Architecture and Gateways

Proposed interplanetary network architectures generally assume that local networks on Earth, the Moon, and Mars are IP-based, while deep-space and interplanetary trunks use BP to interconnect these IP "islands" (12). IP/BP gateways terminate local IP sessions, encapsulate application-layer traffic (for example HTTP or SMTP) into bundle payloads, and forward those bundles over scheduled BP contacts so that remote gateways can reconstruct the original IP traffic within their own local environments (12). This architecture enables reuse of existing Internet applications without requiring native end-to-end IP connectivity across long-delay space links (12).

1.5.2 Naming, DNS, and BP Node Addressing

To integrate BP-based interplanetary connectivity with name resolution, an interplanetary DNS model has been proposed in which each planetary body operates its own DNS root system and associated top-level domains (13). In this model, DNS records for those TLDs resolve primarily to local IP/BP gateways rather than directly reachable off-world IP addresses, and can include resource records that map domain names to BP node identifiers as well as MX records pointing to interplanetary mail gateways (13). DNSSEC can still be applied with distinct trust anchors per world, while most DNS traffic remains local and only application data is carried over BP between planets (13).

1.5.3 Interplanetary Email over the Bundle Protocol

Email is a prominent application driving the deployment of DTN and BP in interplanetary contexts, where entire SMTP exchanges or packeted email messages are encapsulated as bundle payloads and transported between IP networks on different planetary bodies (12, 14). One proposal describes several deployment models where Mail Transfer Agents (MTAs) cooperate with Bundle Protocol Agents

(BPAs) to move email between Earth and remote BP nodes, including cases in which the remote node has only BP connectivity and relies on a local shim to provide SMTP semantics to users or applications (12). A complementary specification defines a concise encapsulation based on Batch SMTP media types and a dedicated BP service number, allowing conventional MTAs at each end to operate unchanged while the interplanetary hop is handled solely via BP (14).

1.6 Interplanetary Overlay Network (ION)

The Interplanetary Overlay Network (ION) is a mature, open-source implementation of the Delay/Disruption-Tolerant Networking (DTN) Bundle Protocol (BP), developed primarily for space communications but also applicable to terrestrial challenged networks (11). ION provides a complete DTN node stack including bundle processing, contact plan management, convergence layer protocols (such as TCPCL and LTP), and administrative tools for monitoring and management (15). It has been deployed in NASA missions, satellite testbeds, and research environments, serving as the de facto reference implementation for BP evaluation and experimentation (11, 15).

1.6.1 System Architecture

ION operates as a distributed daemon-based system where the primary `iond` process manages the global DTN node state, bundle storage, and contact scheduling (11). Bundles are persisted in a SQLite database for reliability across restarts, with separate tables tracking bundle metadata, custody records, and delivery status to support proactive custody transfer and retransmission (15). The architecture separates core BP logic from convergence layers via a plugin model, allowing ION to adapt to diverse underlying links including UDP, TCP, LTP, and space data link protocols without modifying the bundle core (11, 16).

1.6.2 Applications and Deployment

ION ships with reference applications demonstrating BP usage, including the Bundle Relay Utility (BRU) for simple store-forward, the DTNperf tool for throughput measurement, and higher-layer protocol shims for email, file transfer, and web services over BP (15). It has been flight-qualified for NASA's Deep Space Network and CubeSat missions, with operational deployments managing terabytes of scientific data across solar system distances (11). The codebase remains actively maintained with contributions from JPL, ESA, and academic partners, supporting both RFC 5050 BPv6 and emerging BPv7 specifications (15, 16).

1.7 Postfix

Postfix is a Mail Transfer Agent (MTA) that handles sending and receiving emails using the Simple Mail Transfer Protocol (SMTP). It is responsible for transferring emails between mail servers and delivering messages to local user mailboxes in a secure and efficient manner.

1.8 Dovecot

Dovecot is a Mail Delivery Agent (MDA) and IMAP/POP3 server that allows users to access and read their emails stored on the mail server. It manages mailbox formats such as Maildir and provides authentication and retrieval services for email clients.

1.9 Thunderbird

Thunderbird is a Mail User Agent (MUA) that provides a graphical interface for users to compose, send, receive, and manage emails. It communicates with Postfix for sending emails via SMTP and with Dovecot for retrieving emails using IMAP or

POP3.

1.10 Virtual Machines (VMs)

Virtual Machines are isolated computing environments that run independent operating systems on a single physical host. In this setup, VMs are used to simulate separate mail servers and clients, enabling realistic testing and experimentation with email system architectures.

1.11 ION DTN

ION-DTN is the reference implementation of Delay-Tolerant Networking protocols by NASA's Jet Propulsion Laboratory, tailored for space communications where standard Internet protocols are unsuitable because of very high latencies, frequent disconnections, and highly asymmetric data rates. ION implements the Bundle Protocol (16), offering store-and-forward message switching for reliable data transport over challenged networks. The protocol uses Contact Graph Routing (CGR) to determine optimal paths for forwarding messages according to expected contact opportunities between nodes in the network, which is particularly useful for interplanetary communications where link schedules are deterministic but discontinuous. ION also supports a variety of convergence layer adapters, including UDP (for testing on Earth), LTP (Licklider Transmission Protocol (17), for deep space communications), and TCP (18), allowing the protocol to be used in a wide range of network environments. The Bundle Security Protocol (19) provides end-to-end authentication and confidentiality for bundles traversing untrusted intermediary nodes. Originally developed for Mars exploration and the Deep Space Network, ION has been successfully tested in space communications and applied to Earth-based scenarios where disruption-tolerant networking is required, such as disaster recovery networks, military communications, and remote sensing.

1.12 BPMail

1.12.1 Introduction

BPMail serves as a bridge between the conventional email and DTN. When a email is sent from the sender using mail servers like postfix, BPMail intercepts it and encapsulates into Bundle payload and transfer it to the DTN via DTPC. Similarly, when a email arrives in the receiver's side, BPMail decompresses the bundle and reconstructs the original MIME message and deliver it to to receiver's mail server.

1.12.2 Integration with Postfix

BPMail must be integrated with Postfix so that emails destined for the remote DTN node are routed through BPMail's transport rather than the default SMTP delivery path. This integration require configuring BPMail both in sender's side and receiver's side so that the DTN transport is handled seamlessly without any changes to how users send or receive email. On the sending side, Postfix was configured with a custom transport map that routes emails destined for the remote mail server through BPMail instead of attempting direct SMTP delivery. A pipe transport entry was added to Postfix's master.cf, which instructs Postfix to pass outgoing emails to bpmailsend via standard input. On the receiving side, a background daemon continuously runs bpmailrecv, which blocks and waits for incoming DTPC payloads. When an email bundle arrives, bpmailrecv reconstructs the original MIME message and pipes it directly to sendmail, which reintroduces the email into the local Postfix mail system. From here, Postfix handles the delivery of the email to the receiver's mailbox.

1.13 Anti-Spam Systems

Anti-spam systems are software tools deployed on mail servers to filter unwanted bulk or malicious email before it reaches recipients. They work by analysing incoming messages against known indicators of spam such as sender IP reputation, message content patterns, and email authentication records and assigning a verdict or score that determines whether a message should be accepted, flagged, or rejected. These tools rely on real-time network queries to external databases and blacklists to make their decisions, making them dependent on low-latency internet connectivity to function effectively.

1.13.1 SpamAssassin

SpamAssassin is a rule-based anti-spam tool that processes incoming email by applying a combination of local content rules, DNS-based blacklist (DNSBL) lookups, and email authentication checks (20). For each message, it queries multiple external services including IP blacklists, domain blacklists, URI blacklists, and sender accreditation services to evaluate the reputation of the sender IP and any URLs found in the message body. The number of DNS queries it issues scales with the content of the email, increasing with the number of URLs present.

1.13.2 Rspamd

Rspamd is a multi-method anti-spam system that performs DNSBL lookups, SPF, DKIM, and DMARC verification, URI blacklist checks, and ASN lookups in parallel for each incoming message (21). It queries a broad set of external DNS-based services per email and runs continuous background monitoring probes against those services independent of per-email processing. It assigns a weighted score based on the rules triggered and takes a configured action such as adding headers, greylisting, or rejecting based on that score.

1.13.3 Vipul's Razor

Vipul's Razor is a collaborative spam-detection tool that computes fuzzy checksums of email body content and queries Cloudmark's centralised catalogue servers over TCP port 2703 to check whether those checksums match previously reported spam (22). It requires two layers of network connectivity: a DNS lookup to resolve Cloudmark server hostnames, followed by a TCP session to one of the catalogue servers. Unlike DNSBL-based tools, Razor does not evaluate sender IP reputation, it focuses entirely on whether the message content itself has been reported as spam by other users.

1.13.4 DCC (Distributed Checksum Clearinghouse)

DCC is a distributed spam-detection system that computes three types of checksums, Body, Fuz1, and Fuz2 from each incoming message and submits them to a network of globally distributed servers over UDP port 6277 (23). Those servers maintain counts of how many recipients have received messages with matching checksums; messages seen by a large number of recipients are classified as bulk mail. Unlike DNSBL tools, DCC does not check sender reputation, and unlike Razor, it does not rely on user-reported spam signatures. It detects spam purely based on volume of distribution.

2 Literature Review

3 | System Design

This chapter describes the testbed used to evaluate anti-spam tools under the one-way delay of a deep-space DTN link. It sets out the node layout, the host model and the reasons for it, the tools under test, the DTN configuration, the delay simulation, and the experimental design.

3.1 System Architecture

The testbed has two networks joined by a relay. Earth uses the 192.168.100.0/24 subnet and Moon uses the 192.168.200.0/24 subnet. The nodes are:

- **Mail1 (Earth).** The Earth mail server. Runs Postfix, Dovecot, and the four anti-spam tools. User account `alice`. Address 192.168.100.10. ION node 1.
- **Space (Relay).** The transit node between the two networks. Runs an ION node and carries the delay and packet-capture configuration. Earth-side interface 192.168.100.1, Moon-side interface 192.168.200.1. ION node 2.
- **Mail2 (Moon).** The Moon mail server. Runs Postfix and Dovecot. User account `bob`. Address 192.168.200.11. ION node 3.
- **Client1 (Earth).** User terminal on the Earth network. Runs Thunderbird and connects to Mail1 over IMAP (port 143) and SMTP submission (port 587, STARTTLS).
- **Client2 (Moon).** User terminal on the Moon network. Runs Thunderbird and

connects to Mail2 the same way.

Mail moves server to server across the DTN link: a message submitted on one mail server is carried as a bundle through the relay to the other. ION runs on all three servers, so the bundle path is node 1 (Earth) to node 2 (relay) to node 3 (Moon). The relay is the only path between the two networks, so all traffic between Earth and Moon passes through it. This is where the one-way link delay is applied, standing in for the Earth–Moon propagation time. Mailboxes use the mbox format, and each user reads mail from their local server. The topology is shown in Figure 3.1.

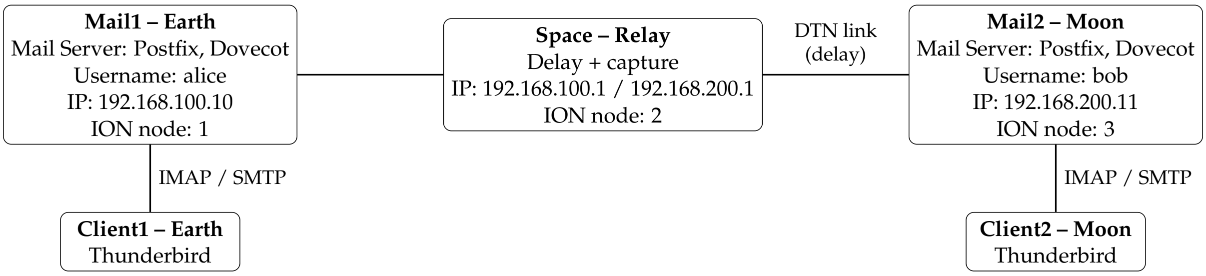

Figure 3.1: Testbed topology. Each client connects only to its local mail server. The relay is the only path between the Earth and Moon networks and is where the one-way link delay is applied.

Data flow (Earth to Moon). The path of a message is shown in Figure 3.2.

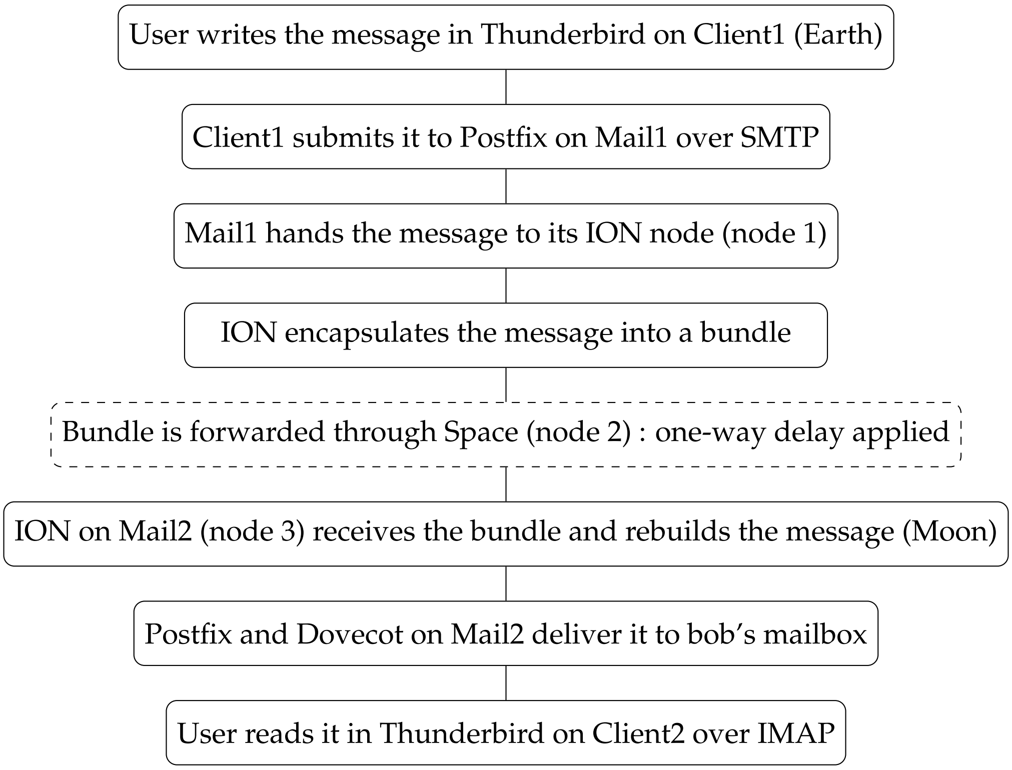

Figure 3.2: Message flow from Earth to Moon.

3.2 Environment and Technology Selection

3.2.1 Virtual machines over containers

The testbed runs on full virtual machines (Ubuntu 22.04 LTS, ARM64, under UTM on macOS). Containers were considered but not used. The anti-spam experiment has three needs that decide this:

- **VMs match the real topology.** The testbed mimics three separate machines — Earth, Space, Moon — connected by a long-delay link, and full VMs naturally model this, whereas containers all share one kernel on one host.
- **Simpler delay setup** Setting up ifb to delay incoming traffic is fiddly and error-prone inside containers, but it's straightforward on a VM where each node has its own kernel and network stack.
- **DTN state stays isolated per node.** Each ION node keeps its own bundle store

and contact schedule, and a full VM ensures the link delay sits on the path itself rather than leaking into a node's own processing.

3.3 Email stack

- **Postfix** was chosen for its clear interface for handing mail to an external program, which is needed to pass messages to the DTN layer.
- **Dovecot** was chosen because it works directly with the mbox format and the local system accounts that Postfix delivers to.
- **Thunderbird** was chosen as a real mail client, so the message traffic in the testbed matches what a normal user would generate.

3.4 DTN stack

- **ION-DTN** was chosen for its deep-space flight heritage and its existing tool for carrying email over the Bundle Protocol, which connects directly to the mail stack.

3.5 Anti-Spam Tool Selection

Four tools were selected, each with a different external dependency:

- **SpamAssassin 3.4.6** — content scoring that depends on DNS blocklist lookups.
- **Rspamd 2.7** — content scoring with a large number of asynchronous DNS lookups.
- **Vipul's Razor 2.84** — collaborative filtering through a central nonce server, reached over TCP on port 2703.
- **DCC 2.3.169** — collaborative filtering through checksum servers, reached over UDP on port 6277.

The four cover the main ways a filter reaches the network: blocklist DNS (SpamAssassin, Rspamd), a TCP request to a central reputation server (Razor), and a UDP request to a checksum server (DCC). Testing one tool would show only one failure path.

3.6 DTN Design

The DTN layer uses ION-DTN 4.1.4 and BPMail. The three servers run ION as nodes on a single `ipn` scheme: Mail1 (Earth) is node 1, Space (relay) is node 2, and Mail2 (Moon) is node 3. Bundles are carried over a UDP convergence layer on port 4556, and the relay forwards them between the two mail nodes, so the path from Earth to Moon is node 1 to node 2 to node 3. The contact plan sets the windows during which the nodes can exchange bundles. BPMail sits above ION on each mail server: it takes a message from Postfix, encapsulates it into a bundle for the destination node, and rebuilds it on arrival. The full configuration, including the SDR settings, the contact plan, and the startup order, is given in the implementation chapter.

3.7 Delay Simulation Design

Delay is applied with netem. The key requirement is that delay is applied to one direction only — to traffic coming from the remote node — and not to the Earth node's own traffic.

A plain netem rule on the relay's interface delays everything that crosses it, including the Earth node's traffic, which would change what is being measured. To avoid this, ingress traffic from the remote side is redirected to an Intermediate Functional Block (ifb) device, and the delay is applied there. The redirect happens before the relay's NAT rewrites the source address, so the rule can match traffic by its true origin. Earth-origin traffic does not pass through this path and stays at normal latency. This is shown in Figure 3.3.

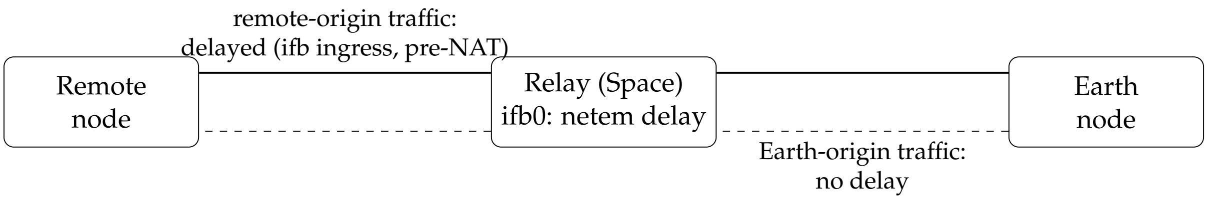

Figure 3.3: Asymmetric delay. Traffic originating from the remote node is delayed on ingress through an ifb device, before NAT rewrites the source. Earth-origin traffic is not delayed.

Delay is the independent variable. The experiments use a representative one-way delay of 1.3 seconds as a deep-space operating point. The testbed allows any value to be set, so the same setup covers the range from short Earth–Moon delays to multi-minute deep-space delays.

3.8 Summary

This chapter set out a three-node testbed for measuring anti-spam tool behaviour under link delay. The design uses full virtual machines for kernel and DNS isolation, a private resolver to give the Earth side a working DNS path, four tools chosen to cover the main external dependencies, and an ifb-based scheme to apply delay to one direction only. The next chapter describes how this design was built and configured, and the chapter after it presents the results.

4 Implementation

Implementation content

5 Results and Discussion

Results

6 Conclusions and Future Work

Conclusions

Bibliography

[1] Kevin Fall. A delay-tolerant network architecture for challenged internets. In *ACM SIGCOMM*, 2003.

[2] Dave Crocker. Internet mail architecture. *RFC 5598*, 2009.

[3] Vinton G. Cerf, Scott C. Burleigh, Adrian J. Hooke, et al. Delay-tolerant networking architecture. *RFC 4838*, 2007.

[4] Aruna Balasubramanian, Brian N. Levine, and Arun Venkataramani. Dtn routing as a resource allocation problem. *ACM SIGCOMM Computer Communication Review*, 2007.

[5] Scott Burleigh et al. Delay-tolerant networking tcp convergence-layer protocol version 4. *RFC 9174*, 2022.

[6] C. C. Sobin et al. A survey of routing and data dissemination in delay tolerant networks. *Ad Hoc Networks*, 2012.

[7] John C. Klensin. Simple mail transfer protocol. *RFC 5321*, 2008.

[8] Mark R. Crispin. Internet message access protocol - version 4rev1. *RFC 3501*, 2003.

[9] Pete Resnick. Internet message format. *RFC 5322*, 2008.

[10] Francesco Buccafurri et al. Secure application email architecture. In *CEUR Workshop Proceedings*, 2023.

[11] Scott C. Burleigh. Interplanetary overlay network: An implementation of the dtn bundle protocol. In *Proceedings of the 2007 4th IEEE Consumer Communications and Networking Conference (CCNC)*, pages 222–226. IEEE, 2007. doi: 10.1109/CCNC.2007.51.

[12] Scott M. Johnson. A method for delivery of smtp messages over bundle protocol networks. Internet-Draft draft-johnson-dtn-interplanetary-smtp-01, Internet Engineering Task Force, 2024. URL `https://datatracker.ietf.org/doc/draft-johnson-dtn-interplanetary-smtp/01/`. Work in progress.

[13] Scott M. Johnson. An interplanetary dns model. Internet-Draft draft-johnson-dtn-interplanetary-dns-04, Internet Engineering Task Force, 2025. URL `https://datatracker.ietf.org/doc/draft-johnson-dtn-interplanetary-dns/04/`. Work in progress.

[14] Marc Blanchet. Encapsulation of email over delay-tolerant networks (dtn) using the bundle protocol. Internet-Draft draft-blanchet-dtn-email-over-bp-06, Internet Engineering Task Force, 2025. URL `https://datatracker.ietf.org/doc/draft-blanchet-dtn-email-over-bp/`. Work in progress.

[15] Jet Propulsion Laboratory. Interplanetary overlay network (ion) – official repository. `https://github.com/InterplanetaryOverlayNetwork/ion`, 2025. Accessed: 2025-12-07.

[16] Scott Burleigh, Kevin Fall, and Edward J. Birrane. Bundle Protocol Version 7. RFC 9171, January 2022. URL `https://www.rfc-editor.org/info/rfc9171`.

[17] Scott C. Burleigh, Stephen Farrell, and Manikantan Ramadas. Licklider Transmission Protocol - Specification. RFC 5326, September 2008. URL `https://www.rfc-editor.org/info/rfc5326`.

[18] Michael Demmer, Joerg Ott, and Simon Perreault. Delay-Tolerant Networking TCP Convergence-Layer Protocol. RFC 7242, June 2014. URL https://www.rfc-editor.org/info/rfc7242.

[19] Edward J. Birrane and Kenneth McKeever. Bundle Protocol Security (BPSec). RFC 9172, January 2022. URL https://www.rfc-editor.org/info/rfc9172.

[20] Apache Software Foundation. Apache SpamAssassin. https://spamassassin.apache.org, 2024. Accessed: 2024.

[21] Vsevolod Stakhov. Rspamd spam filtering system. https://rspamd.com/doc/, 2024. Accessed: 2024.

[22] Vipul Ved Prakash and Chad Norwood. Razor-agents: Collection of files for checking, reporting, and revoking spam. Ubuntu Manpages. https://manpages.ubuntu.com/manpages/noble/man5/razor-agents.5.html, 2024. Accessed: 2024.

[23] Rhyolite Software. Distributed checksum clearinghouse (DCC). https://www.dcc-servers.net/dcc/, 2024. Accessed: 2024.